# lysA-I (Q88L58 / PP_2077): Diaminopimelate Decarboxylase of *Pseudomonas putida* KT2440

**Gene:** lysA-I (synonym *lysA*; ordered locus PP_2077) **Protein:** Diaminopimelate decarboxylase (DAPDC / DAP decarboxylase) **UniProt:** Q88L58 · **EC:** 4.1.1.20 · **Organism:** *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / KT2440), PSEPK

## Summary

**lysA-I encodes diaminopimelate decarboxylase (DAPDC; EC 4.1.1.20), the pyridoxal-5′-phosphate (PLP)-dependent enzyme that catalyzes the terminal, committed, and irreversible step of L-lysine biosynthesis: meso-2,6-diaminopimelate + $H^+$ → L-lysine + $CO_2$ (RHEA:15101).** This identification is unambiguous and internally consistent: the target gene symbol, UniProt annotation, enzyme classification, protein family (Orn/Lys/Arg decarboxylase class-II, i.e., fold-type III / alanine-racemase fold), and conserved catalytic domains all converge on this single biochemical function. The verification steps mandated by the research brief are satisfied — the gene symbol lysA-I matches the DAP decarboxylase description, the organism is correctly *P. putida* KT2440, and the family/domain architecture aligns with the DAPDC literature. No competing gene identity was encountered.

The enzyme is a soluble, cytoplasmic, obligate homodimer with no signal peptide and no membrane-spanning segment. It performs its chemistry in the cytosol, where it sits at a pivotal **metabolic branch point**: its substrate, meso-diaminopimelate (meso-DAP), is simultaneously a direct cross-linking residue of Gram-negative peptidoglycan and the immediate precursor of free L-lysine. By decarboxylating meso-DAP, DAPDC partitions carbon and nitrogen flux between cell-wall construction (retaining meso-DAP) and protein synthesis (producing L-lysine). It is the single convergent terminal enzyme onto which two annotated meso-DAP-producing routes in *P. putida* — the succinyl-DAP pathway (KEGG M00016) and the DAP-aminotransferase pathway (KEGG M00527) — both feed.

Mechanistically, DAPDC is a fold-type III PLP decarboxylase that is highly specific for meso-DAP. Specificity is achieved by recognition of the L-configured stereocenter of the substrate combined with a system of ionic "molecular rulers" — two active-site arginines (Arg262/Arg298 in Q88L58) — that gauge substrate length. Catalysis proceeds by a **two-base mechanism** involving the PLP internal aldimine to Lys45 and a catalytic Cys332 proton donor, and the reaction runs with **net inversion of configuration** at the reacting carbon, a hallmark that distinguishes meso-DAP decarboxylase from other α-amino-acid decarboxylases. An AlphaFold model (global pLDDT 95.1) independently reproduces the canonical two-domain architecture and places all functional residues at very-high confidence, with a composite active site assembled across the dimer interface. Because the DAP/lysine pathway is absent in mammals and is essential in bacteria where tested, DAPDC is a validated, selective antibacterial target — although in *P. putida* a second, ~48%-identical paralog (lysA-II / PP_5227) may provide functional redundancy.

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## Key Findings

### F001 — Core identity and reaction

Q88L58 (lysA-I / PP_2077) is a PLP-dependent diaminopimelate decarboxylase catalyzing the final step of lysine biosynthesis. UniProtKB assigns EC 4.1.1.20 and the catalytic reaction **meso-2,6-diaminopimelate + $H^+$ → L-lysine + $CO_2$** (RHEA:15101). The cofactor pyridoxal-5′-phosphate is bound as an internal aldimine (Schiff base) to **Lys45** (annotated feature "N6-(pyridoxal phosphate)lysine" at position 45). The mature protein is 410 amino acids and assembles as a **homodimer**. It occupies the terminal position in the diaminopimelate (DAP) pathway of L-lysine biosynthesis, formally "L-lysine from DL-2,6-diaminopimelate: step 1/1." Gene Ontology annotations are fully consistent: GO:0008836 (diaminopimelate decarboxylase activity), GO:0030170 (PLP binding), and GO:0009089 (L-lysine biosynthesis via DAP). The active-site lysine/proton-donor and substrate/PLP-binding residues are annotated at positions 218, 259, 262, 298, 302, 332/333, 367. The protein belongs to the Orn/Lys/Arg decarboxylase class-II family (fold-type III, alanine-racemase fold). This is the anchoring finding: the identity of the gene product as DAPDC is unambiguous and self-consistent across reaction, cofactor, pathway, and family annotations.

## F002 — Two divergent DAPDC paralogs in *P. putida* KT2440

*P. putida* KT2440 encodes **two** divergent DAP decarboxylase paralogs. A UniProt taxonomy search (taxid 160488, EC 4.1.1.20) returns exactly two entries: **Q88L58 = lysA-I / PP_2077 (410 aa)** and **Q88CF4 = lysA-II / PP_5227 (415 aa)**, both annotated as diaminopimelate decarboxylase. A Needleman–Wunsch global alignment of the two sequences yields 174/364 aligned identical positions = **47.8% pairwise identity**. Both retain the class-II Orn/Lys/Arg decarboxylase fold and the DAP-decarboxylase family signatures. The presence of two moderately divergent paralogs raises the possibility of functional redundancy, differential regulation, or subtle substrate/kinetic specialization — an important caveat for any inference about essentiality of lysA-I alone in this organism.

## F003 — Mechanism and substrate specificity: fold-type III PLP decarboxylase acting with inversion

DAPDC recognizes the L-stereocenter of meso-DAP and decarboxylates with inversion of configuration. Cocrystal structures of *Methanococcus jannaschii* DAPDC with a substrate analog (azelaic acid, Ki = 89 μM) and with the L-lysine product (2.0–2.6 Å) demonstrated that **substrate specificity arises from recognition of the L-chiral center of diaminopimelate and a system of ionic "molecular rulers" that dictate substrate length** (PMID: 12429091, Ray et al. 2002). meso-DAP possesses one D- and one L-configured stereocenter; DAPDC removes $CO_2$ from the D-configured carbon while recognizing the L-stereocenter, and the reaction proceeds with **inversion of configuration** at the reacting carbon. This stereochemical course was established by $^{2}$H-labeling/NMR studies for the *Bacillus sphaericus* and wheat-germ meso-DAP decarboxylases (PMID: 3927976, Kelland et al. 1985), and it contrasts with the *retention* of configuration seen with many other α-amino-acid decarboxylases — a mechanistically diagnostic distinction. In Q88L58 the corresponding functional features are the PLP internal aldimine to Lys45, the catalytic proton donor at the active site, and substrate/PLP-binding residues at 218/259/262/298/302/333/367.

> "This PLP-dependent enzyme is responsible for the final step of L-lysine biosynthesis in bacteria" — PMID: 12429091

## F004 — Branch-point role and validated antibacterial target

The lysA-I product, L-lysine, sits at the branch point between protein synthesis and peptidoglycan/cell-wall metabolism, and the DAP pathway is an essential, validated antibacterial target. DAPDC catalyzes the terminal step converting meso-DAP → L-lysine (UniProt: step 1/1). Crucially, **meso-DAP itself is a direct cross-linking residue of Gram-negative peptidoglycan**, so DAPDC governs the flux split between a cell-wall precursor (meso-DAP retained) and free L-lysine (committed to protein synthesis). As noted by Zheng et al. 2022, "*Diaminopimelic acid (DAP) is a unique component of the cell wall of Gram-negative bacteria*" (PMID: 36040174). Genetic essentiality of the terminal enzyme has been demonstrated in other bacteria: conditional knockdown of *lysA* in *Mycobacterium tuberculosis* confirmed its essentiality for growth (PMID: 22840772, Abrahams et al. 2012). Because mammals lack the DAP pathway entirely (they obtain lysine from the diet), DAPDC has **no human ortholog**, making it an attractive antibacterial and herbicide target. Regarding localization, Q88L58 has no signal peptide and no membrane-spanning segment (maximum Kyte–Doolittle 19-residue hydropathy = 1.11, well below the ~1.6 transmembrane threshold), consistent with a **soluble cytoplasmic enzyme** with homodimeric quaternary structure.

> "The essentiality of panC and lysA" — PMID: 22840772

## F005 — Two-base active site: PLP–Lys45, catalytic Cys332, and Arg262/Arg298 molecular rulers

UniProt residue-level features for Q88L58 (verified against the sequence) define the catalytic machinery: the **modified residue Lys45** carries the N6-(pyridoxal-phosphate)lysine internal aldimine; the **active site Cys332** is the proton donor (in the motif GPLCESGD); and substrate/PLP-binding residues include Glu259, Arg262, Arg298, Tyr302, Glu333, and Tyr367. The two active-site arginines, **Arg262 and Arg298**, correspond to the ionic "molecular rulers" that recognize the distal carboxylate/amino groups of meso-DAP and thereby dictate substrate length — the same feature demonstrated crystallographically in the *M. jannaschii* enzyme (PMID: 12429091). The catalytic cysteine sits on the opposite face of the PLP–substrate aldimine from Lys45, providing the **second base** required for decarboxylation with net inversion of configuration. Notably, the catalytic-Cys motif GP[L/I]C[E][S/T]GD is conserved in **both** *P. putida* paralogs — lysA-I "GPLCESGD" (Cys332) and lysA-II "GPICETGD" (aligned Cys) — indicating both paralogs preserve the canonical two-base mechanism. Tyr367 is a cross-subunit residue, consistent with the obligate homodimer in which the active site is completed by the partner monomer.

## F006 — Divergent LysR regulator (PP_2078) implies local transcriptional control

lysA-I (PP_2077) is genomically divergent from a dedicated LysR-family transcriptional regulator (PP_2078). KEGG genomic coordinates (*P. putida* KT2440) place **PP_2077/lysA-I on the complement strand, complement(2364209..2365441)**, immediately adjacent to **PP_2078/lysR (LysR-family transcriptional activator) on the forward strand, 2365565..2366470**. The two genes are **divergently transcribed**, with their 5′ ends facing each other across a ~124 bp intergenic region (2365442–2365564) — the canonical architecture of a LysR-type transcriptional regulator (LTTR) controlling a divergent target gene. The flanking genes PP_2075/PP_2076 (complement strand, "unknown function") are separated from lysA-I by a ~330 bp gap, arguing against a single operon with DAP-synthesis genes; lysA-I appears to be its own transcriptional unit. KEGG orthology K01586 (EC 4.1.1.20); Pfam motifs Orn_Arg_deC_N, Orn_DAP_Arg_deC, Ala_racemase_N. This genomic architecture suggests local, likely lysine/DAP-responsive transcriptional control of the terminal biosynthetic step.

## F007 — Convergent terminal enzyme for two meso-DAP routes

In *P. putida* KT2440, lysA-I acts as the common terminal enzyme for two annotated meso-DAP-producing routes. KEGG maps PP_2077 (K01586) to the Lysine biosynthesis pathway ppu00300 and to two lysine-biosynthesis modules: **M00016 (Lysine biosynthesis, succinyl-DAP pathway, aspartate ⇒ lysine)** and **M00527 (Lysine biosynthesis, DAP-aminotransferase pathway, aspartate ⇒ lysine)**. Both modules share the early aspartate → tetrahydrodipicolinate steps and converge on meso-2,6-diaminopimelate, which DAPDC (LysA) decarboxylates to L-lysine. Additional KEGG pathway memberships include D-amino acid metabolism (ppu00470) and biosynthesis of amino acids (ppu01230). DAPDC is therefore the single convergent funnel through which two upstream routes reach lysine.

## F008 — AlphaFold confirms the two-domain fold and composite dimer-interface active site

The AlphaFold model of Q88L58 (AF-Q88L58-F1, v6) confirms a high-confidence two-domain DAPDC fold. The **global pLDDT** = **95.1**, with 89% of residues at very-high confidence (>90) and 97% above 70. All annotated functional residues are modeled at very-high confidence: Lys45 =

98.4, Glu259 = 97.5, Arg262 = 96.7, Arg298 = 97.5, Tyr302 = 93.5, Cys332 = 97.6, Glu333 = 97.6, Tyr367 = 98.2. Within the monomer, the PLP-binding Lys45 clusters spatially with Glu259 (Cα–Cα 11.0 Å), Arg262 (9.3 Å), and Arg298 (13.9 Å), forming a contiguous substrate/cofactor pocket, whereas the catalytic Cys332 lies 27.7 Å from Lys45 and Tyr367 is a cross-subunit residue — geometry consistent with DAPDC's known **composite active site assembled across the homodimer interface**. The N-terminal region (residues 1–290, mean pLDDT 94.3) corresponds to the $(\beta/\alpha)_8$ TIM-barrel-like PLP-binding domain, and the C-terminal region (300–410, 96.9) to the β-sandwich domain — the canonical two-domain fold-type III architecture (Pfam Orn_Arg_deC_N + Orn_DAP_Arg_deC; InterPro IPR009006/IPR002986).

## F009 — Authoritative review validates the dual metabolic destination and target status

An authoritative review confirms that lysine biosynthesis provides both L-lysine (protein synthesis) and meso-DAP (peptidoglycan), validating DAPDC's branch-point role and antibacterial-target status. Hutton, Southwood & Turner (2003), *Inhibitors of lysine biosynthesis as antibacterial agents* (PMID: 12570844), state that the bacterial lysine biosynthetic pathway "provides both lysine for protein synthesis and meso-diaminopimelate for construction of the bacterial peptidoglycan cell wall," and has "come under increased scrutiny as a target for novel antibacterial agents." DAPDC (LysA) catalyzes the terminal meso-DAP → L-lysine step, sitting exactly at this branch. Combined with the mammalian absence of the pathway and demonstrated bacterial essentiality of *lysA* (PMID: 22840772), this establishes DAPDC as a selective antibacterial target.

> "it provides both lysine for protein synthesis and meso-diaminopimelate for construction of the bacterial peptidoglycan cell wall" — PMID: 12570844
>
> "Bacterial biosynthesis of lysine has come under increased scrutiny as a target for novel antibacterial agents" — PMID: 12570844

# Mechanistic Model and Interpretation

## The catalyzed reaction

```
meso-2,6-diaminopimelate  +  H+   ──DAPDC (PLP)──►   L-lysine  +  CO2
(RHEA:15101, EC 4.1.1.20)
```

DAPDC is a **PLP-dependent, fold-type III (alanine-racemase-fold)** decarboxylase. The reaction is the **terminal, committed, irreversible** step of the DAP pathway and is the sole biochemical route in bacteria for generating free L-lysine from the DAP intermediate pool.

## Catalytic cycle (two-base mechanism)

```
1. Resting state:   PLP = Nε-Lys45  (internal aldimine / Schiff base)
2. Transaldimination: meso-DAP amino group displaces Lys45
                      → external aldimine (PLP = substrate)
3. Decarboxylation: CO2 lost from the D-carboxyl; PLP acts as
                    electron sink → quinonoid intermediate
                    (Arg262 / Arg298 molecular rulers clamp the
                     distal carboxylate & amino group, enforcing
                     meso-DAP length/geometry)
4. Reprotonation:   catalytic Cys332 delivers H+ to the OPPOSITE
                    face → NET INVERSION of configuration
5. Product release: L-lysine released; Lys45 re-forms internal aldimine
```

The two-base arrangement — Lys45 on one face (forms the aldimine) and Cys332 on the opposite face (reprotonates the quinonoid) — is what produces the diagnostic **inversion of configuration** observed by NMR/$^2$H-labeling in orthologous enzymes (PMID: 3927976). The "molecular rulers" (Arg262/Arg298) explain the strict specificity for meso-DAP over shorter diamino acids (PMID: 12429091).

## Quaternary structure and active-site geometry

The enzyme is an **obligate homodimer**. Within a single monomer the PLP pocket (Lys45, Glu259, Arg262, Arg298) is contiguous, but the catalytic Cys332 (~28 Å away in the monomer model) and the cross-subunit residue Tyr367 indicate that a **complete active site is built across the dimer interface**, requiring both protomers. AlphaFold (pLDDT 95) independently reproduces the two-domain fold and the spatial clustering of functional residues, corroborating the crystallographic paradigm without an experimental *P. putida* structure.

## Position in metabolism — the branch point

```
        aspartate
           │  (shared early steps → tetrahydrodipicolinate)
           ▼
   ┌────────────────────────────────┐
   │  succinyl-DAP route (M00016)   │──┐
   ├────────────────────────────────┤  │
   │  DAP-aminotransferase (M00527) │──┤ converge on
   └────────────────────────────────┘  ▼
                      meso-2,6-DIAMINOPIMELATE  ◄── peptidoglycan
                                │                    cross-linking
                        lysA-I  │ (DAPDC, PP_2077)   (meso-DAP retained)
                                ▼
                           L-LYSINE  ──► protein synthesis
```

DAPDC is the **convergent funnel** for both annotated meso-DAP routes (F007) and the **valve** that decides whether meso-DAP is committed to lysine or retained for cell-wall cross-linking (F004, F009). Its genomic pairing with a divergent LysR regulator (PP_2078; F006) suggests this valve is under local transcriptional control, plausibly responsive to lysine/DAP status.

## Localization

| Property | Value | Interpretation |
|---|---|---|
| Signal peptide | Absent | Not secreted |
| Transmembrane segments | 0 (max hydropathy 1.11 < 1.6) | Not membrane-associated |
| Quaternary structure | Homodimer | Composite active site |
| Compartment | Cytoplasm | Soluble cytosolic enzyme |

## Paralog comparison

| Feature | lysA-I (Q88L58 / PP_2077) | lysA-II (Q88CF4 / PP_5227) |
|---|---|---|
| Length | 410 aa | 415 aa |
| Annotation | Diaminopimelate decarboxylase | Diaminopimelate decarboxylase |
| Catalytic-Cys motif | GPLCESGD | GPICETGD |
| Pairwise identity | — | 47.8% to lysA-I |
| Fold / family | Class-II Orn/Lys/Arg deCase (fold-type III) | Same |

Both paralogs preserve the two-base catalytic motif, so functional redundancy is plausible; whether they differ in regulation, expression, or fine kinetics is unresolved.

## Evidence Base

| PMID | Title (abbrev.) | Role in this report |
|---|---|---|
| 12429091 | *Cocrystal structures of diaminopimelate decarboxylase* (Ray et al. 2002) | Primary structural evidence for L-stereocenter recognition and "molecular ruler" substrate-length specificity; confirms PLP dependence and terminal-step role. Supports F003, F005. |
| 3927976 | *Stereochemistry of lysine formation by meso-DAP decarboxylase (wheat germ)* (Kelland et al. 1985) | $^2$H-NMR evidence establishing **inversion of configuration** — mechanistic hallmark. Supports F003. |
| 22840772 | *Pathway-selective sensitization of M. tuberculosis* (Abrahams et al. 2012) | Demonstrates genetic **essentiality of lysA** in a bacterium. Supports F004, F009. |
| 12570844 | *Inhibitors of lysine biosynthesis as antibacterial agents* (Hutton et al. 2003) | Authoritative review: dual destination (lysine + meso-DAP for peptidoglycan); antibacterial-target status. Supports F004, F009. |
| 36040174 | *Diaminopimelic Acid Metabolism* (Zheng et al. 2022) | meso-DAP as a unique Gram-negative cell-wall component, defining the branch point. Supports F004. |
| 8206742 | *Regulation of DHDPS and DAP decarboxylase in B. stearothermophilus* | Context on feedback inhibition of DAP decarboxylase by L-lysine in a related bacterium (background). |

**Database and computational evidence:** UniProtKB Q88L58 (reaction, cofactor, residue-level features, GO terms, family); KEGG *P. putida* KT2440 (K01586, ppu00300, modules M00016/M00527, genomic coordinates); AlphaFold DB AF-Q88L58-F1 (pLDDT 95.1); Pfam/InterPro domain assignments; Needleman–Wunsch paralog alignment (47.8% identity); Kyte–Doolittle hydropathy analysis.

**Note on organism specificity:** the mechanistic and structural details (F003, F005) are drawn from orthologous DAPDC enzymes (*M. jannaschii*, wheat germ, *B. sphaericus*), transferred to Q88L58 by sequence/family homology and the conservation of the exact catalytic residues (Lys45, Cys332, Arg262/Arg298). No experimental study of the *P. putida* KT2440 lysA-I protein itself was found; conclusions about it rest on annotation, homology, and AlphaFold modeling.

## Limitations and Knowledge Gaps

1. **No direct experimental characterization of Q88L58.** All functional claims for the *P. putida* KT2440 protein derive from UniProt/KEGG annotation, homology to structurally characterized orthologs, and AlphaFold modeling. There is no published enzymology (kinetics, Km/kcat for meso-DAP), no crystal structure, and no gene-knockout phenotype specifically for lysA-I.
2. **Paralog redundancy is unresolved.** *P. putida* KT2440 has two DAPDC paralogs (~48% identical). Whether lysA-I is individually essential — or whether lysA-II can compensate — is unknown. Essentiality was demonstrated for *lysA* in *M. tuberculosis* (single-copy), which does not necessarily transfer to a two-paralog organism.
3. **Regulation is inferred, not demonstrated.** The divergent LysR regulator (PP_2078) implies local transcriptional control, but no experiment defines the effector, the operator, or whether PP_2078 activates or represses lysA-I. Feedback inhibition of DAP decarboxylase by L-lysine is documented in *B. stearothermophilus* but not verified for *P. putida* lysA-I.
4. **Stereochemistry/inversion is transferred from orthologs.** The inversion-of-configuration mechanism was established in wheat-germ and *B. sphaericus* enzymes; it is assumed conserved in lysA-I based on identical catalytic-residue architecture but not directly measured.
5. **Which upstream route dominates in vivo is unknown.** KEGG lists both the succinyl-DAP and DAP-aminotransferase routes; the relative in-vivo flux through each in *P. putida* KT2440 has not been quantified.
6. **AlphaFold is a monomer model.** The composite dimer-interface active site is inferred from residue geometry and orthologous structures; the dimer itself was not modeled or experimentally confirmed for this protein.

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## Proposed Follow-up Experiments / Actions

1. **Recombinant enzymology.** Express and purify Q88L58, reconstitute with PLP, and measure steady-state kinetics (Km, kcat) with meso-DAP; test specificity against LL-DAP, DD-DAP, and shorter diamino acids to confirm the "molecular ruler" prediction.

2. **Active-site mutagenesis.** Generate K45A, C332A/C332S, R262A, and R298A variants to validate the two-base mechanism and the ionic-ruler roles; expect loss of activity for K45/C332 and relaxed length specificity for R262/R298.
3. **Stereochemical assay.** Use $^2H_2O$ incubation with chiral NMR (as in PMID: 3927976) to directly confirm inversion of configuration for the *P. putida* enzyme.
4. **Genetics of redundancy.** Construct single (ΔlysA-I, ΔlysA-II) and double knockouts in KT2440; test lysine auxotrophy and cross-complementation to establish individual vs. combined essentiality.
5. **Regulation.** Build a lysA-I promoter–reporter fusion; delete/overexpress PP_2078 (LysR) and vary lysine availability to define the regulatory logic and effector.
6. **Structure.** Determine an experimental crystal or cryo-EM structure of the homodimer, ideally with PLP and a substrate analog (e.g., azelaic acid), to confirm the cross-subunit composite active site predicted by AlphaFold.
7. **Feedback inhibition.** Test L-lysine (and analogs such as S-2-aminoethyl-cysteine) as inhibitors of purified DAPDC, paralleling the *B. stearothermophilus* observations (PMID: 8206742).
8. **Inhibitor / target validation.** Given the mammalian absence of the pathway, screen DAPDC-directed inhibitors against both paralogs for antibacterial-lead potential.

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## Conclusion

lysA-I (Q88L58 / PP_2077) of *Pseudomonas putida* KT2440 is **diaminopimelate decarboxylase (DAPDC, EC 4.1.1.20)** — a PLP-dependent, fold-type III, cytoplasmic homodimeric enzyme that catalyzes the final, committed, irreversible step of L-lysine biosynthesis (meso-2,6-diaminopimelate → L-lysine + $CO_2$). It is highly specific for meso-DAP via L-stereocenter recognition and Arg262/Arg298 molecular rulers, uses a two-base (PLP-Lys45 / Cys332) mechanism proceeding with inversion of configuration, sits at the meso-DAP branch point between protein synthesis and peptidoglycan, is the convergent terminal enzyme for two upstream DAP routes, is under likely LysR (PP_2078) control, and — being absent in mammals — is a validated antibacterial target, with a ~48%-identical paralog (lysA-II / PP_5227) as a possible redundancy factor.